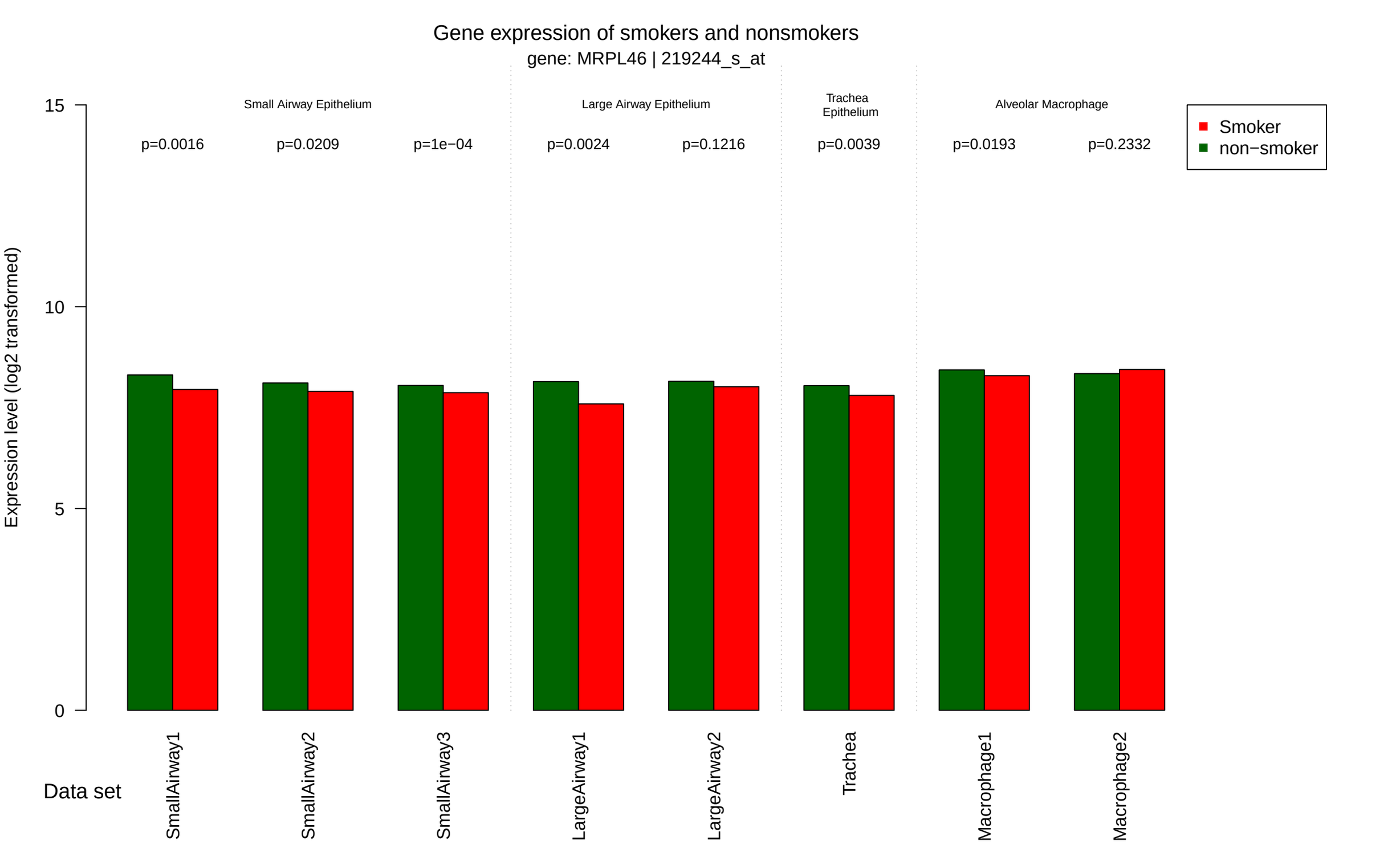
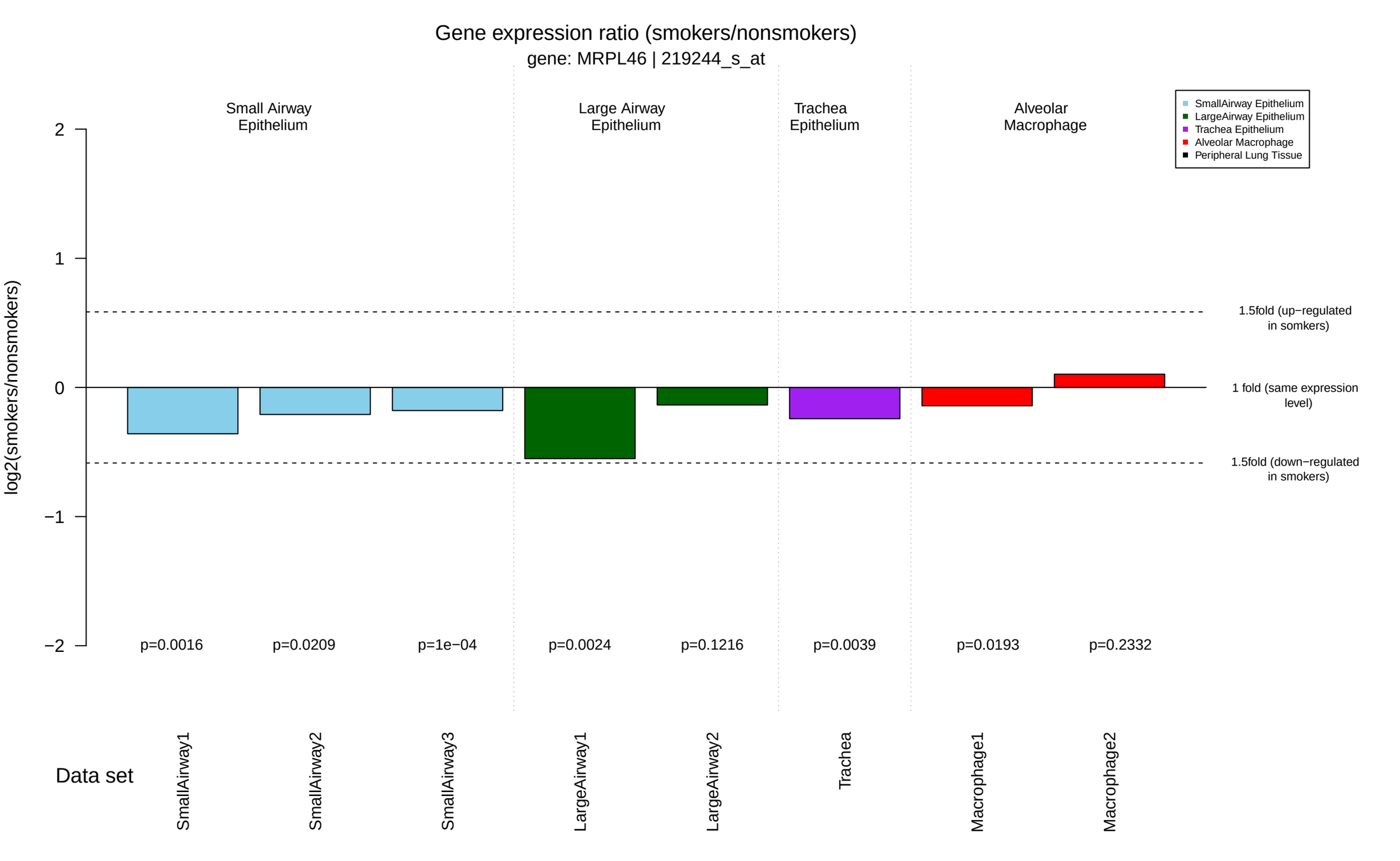
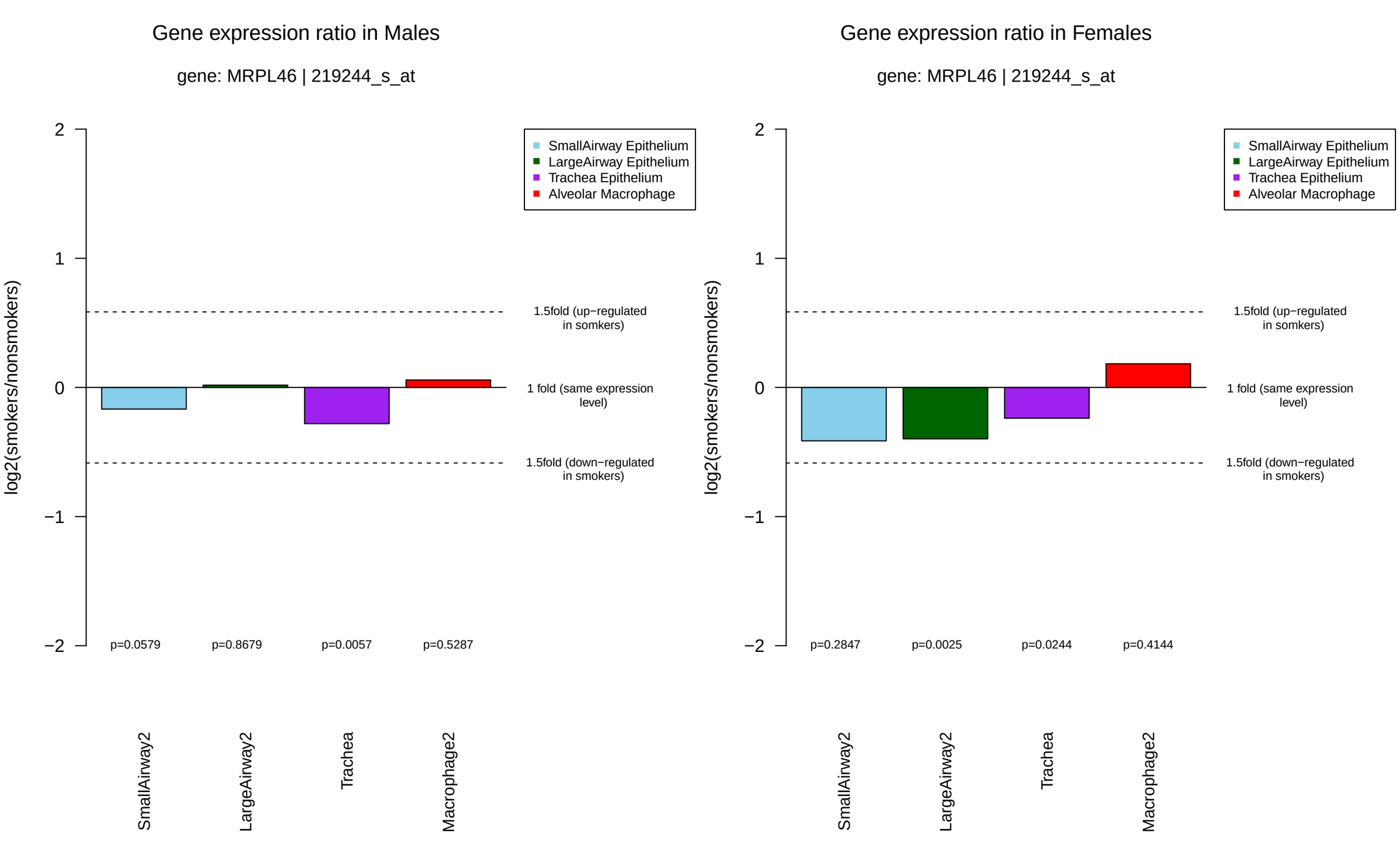
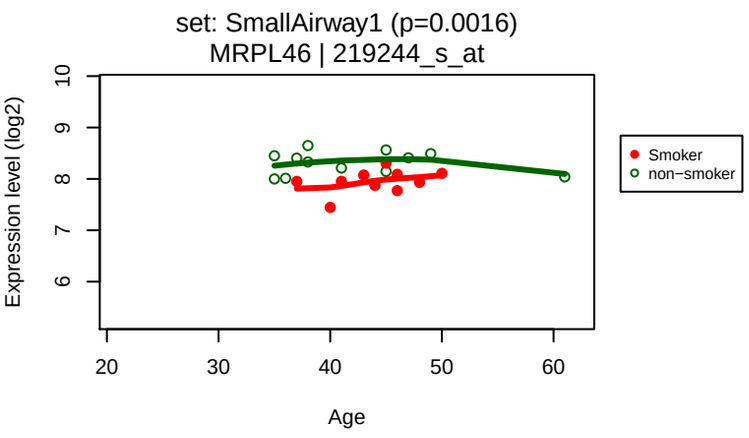
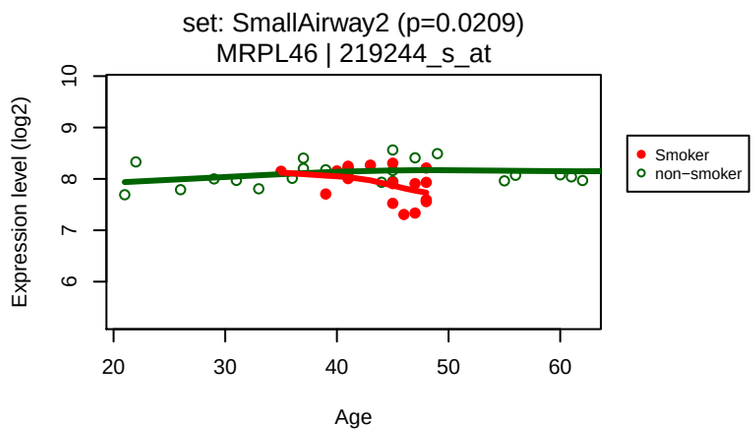
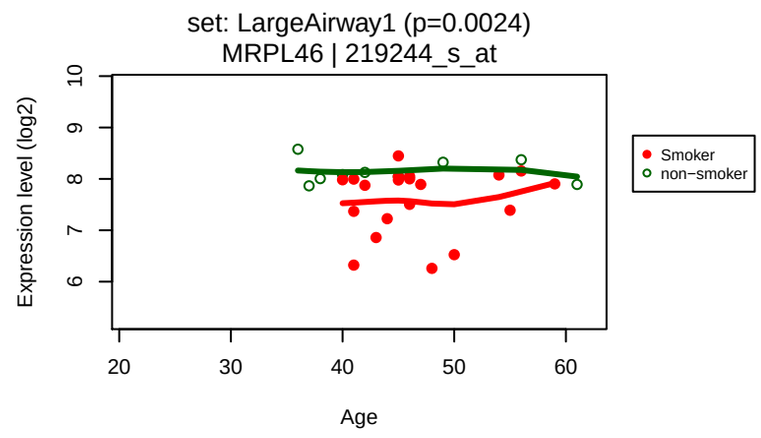

Gene expression of smokers and nonsmokers
gene: MRPL46 | 219244_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0016
p=0.0209
p=1e−04
p=0.0024
p=0.1216
p=0.0039
p=0.0193
p=0.2332
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

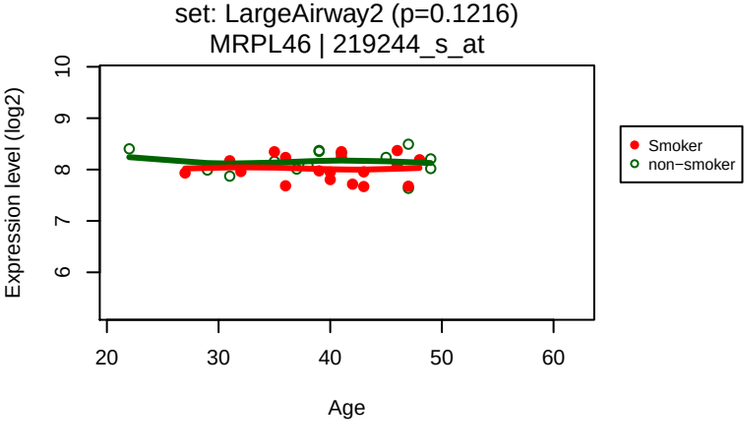

Gene expression ratio (smokers/nonsmokers)
gene: MRPL46 | 219244_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0016
p=0.0209
p=1e−04
p=0.0024
p=0.1216
p=0.0039
p=0.0193
p=0.2332
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

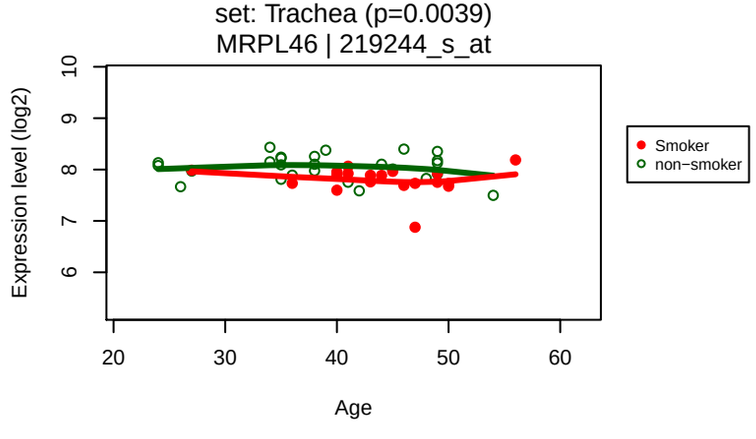

Gene expression ratio in Males
gene: MRPL46 | 219244_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0579
p=0.8679
p=0.0057
p=0.5287
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRPL46 | 219244_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2847
p=0.0025
p=0.0244
p=0.4144
SmallAirway2
LargeAirway2
Trachea
Macrophage2

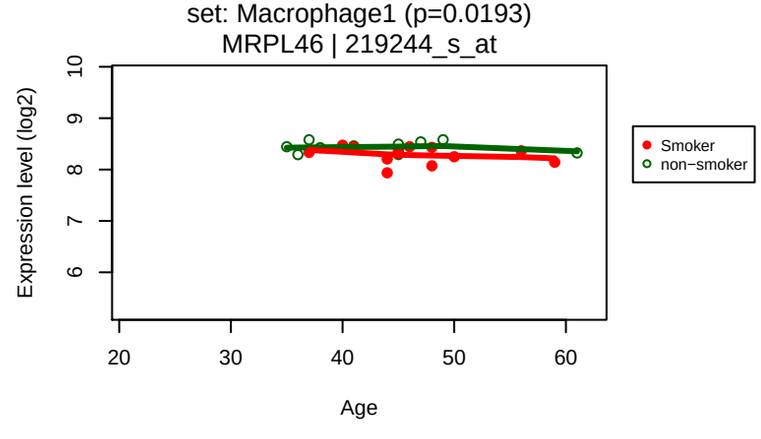

set: SmallAirway1 (p=0.0016)
MRPL46 | 219244_s_at
Expression level (log2)
Age
Smoker
non−smoker

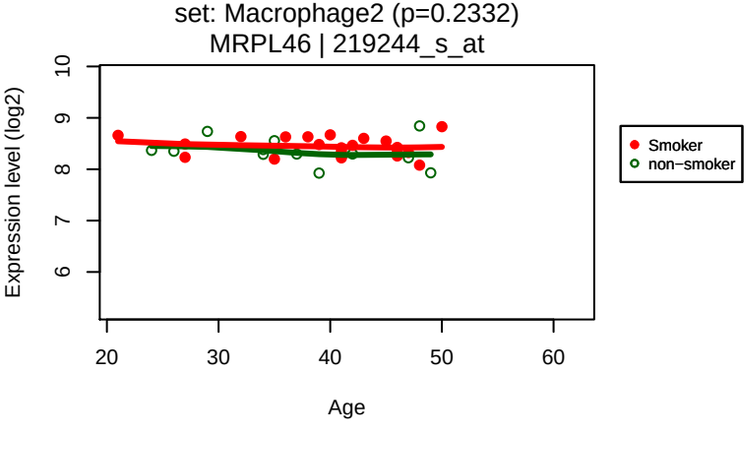

set: SmallAirway2 (p=0.0209)
MRPL46 | 219244_s_at
Expression level (log2)
Age
Smoker
non−smoker

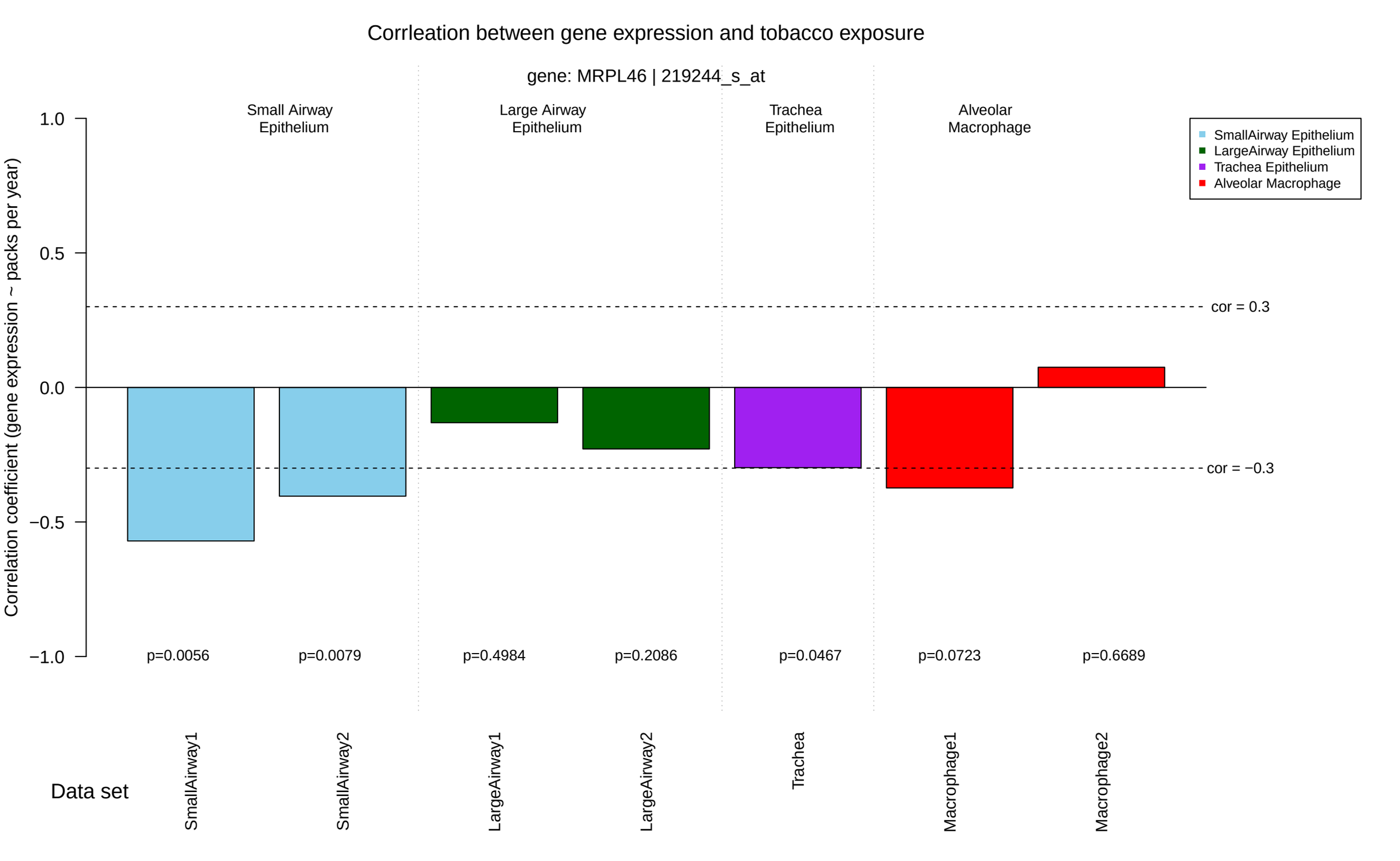

set: LargeAirway1 (p=0.0024)
MRPL46 | 219244_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1216)
MRPL46 | 219244_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0039)
MRPL46 | 219244_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0193)
MRPL46 | 219244_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2332)
MRPL46 | 219244_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRPL46 | 219244_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0056
p=0.0079
p=0.4984
p=0.2086
p=0.0467
p=0.0723
p=0.6689
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2